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MATH 245A

Discussion 0
Due:

Throughout, $m^*(A) \coloneqq \inf\left\{\sum_{i=1}^{\infty} |b_i - a_i| : A \subseteq \bigcup_{i=1}^{\infty}(a_i, b_i)\right\}$ for $A \subseteq \mathbb{R}$, and for measurable A we write $m(A) \coloneqq m^*(A)$ (Lebesgue measure).

Problem 1.1

If $U \subseteq \mathbb{R}$ is open, prove U is a countable (finite or infinite) union of disjoint open intervals $U = \bigcup_i (a_i, b_i)$, where $a_i = -\infty$, $b_i = +\infty$ are allowed.

> *Solution.* U can be split into connected components (each of which are open intervals). These are countable because $\mathbb{R}$ is seperable and $\mathbb{Q}$ is countable dense.

Problem 1.2

Prove that m^* is an outer measure.

> *Solution.* Nonnegativity is obvious. Suppose $A \subseteq B$. Suppose
>
> $$B \subseteq \bigcup_{i=1}^{\infty}(a_i, b_i) \Rightarrow A \subseteq B \subseteq \bigcup_{i=1}^{\infty}(a_i, b_i) \Rightarrow m^*(A) \leq \sum_{k=1}^{\infty}|b_i - a_i|.$$
>
> This is true for all such covers, thus $m^*(A) \leq m^*(B)$. Thus outer measure is monotonic with respect to containment. Furthermore we need to show subadditivity:
>
> $$m^*(A \cup B) \leq m^*(A) + m^*(B)$$
>
> Basically you take any pair of covers of A and B and $m^*(A \cup B)$ is a lower bound for the sums. This is true for all arbitrary pairs and thus must also be true for the infimums.

Problem P1

Does there exist a measurable $A \subseteq \mathbb{R}$ with $m(A) > 0$ and $\varepsilon > 0$ such that for every interval (a, b),

$$m(A \cap (a, b)) \leq (1 - \varepsilon)\ |b - a|?$$

> *Solution.* Assume the contrary that such a A and $\varepsilon > 0$ exists. We will reduce to finite measure by intersecting A with a compact set.
>
> Since A is measurable and $m(A) > 0$. Let $\{(a_i, b_i)\}_{i=1}^{\infty}$ be any collection of open intervals such that
>
> $$A \subseteq \bigcup_{k=1}^{\infty}(a_k, b_k) \Rightarrow A \subseteq \bigcup_{k=1}^{\infty}(A \cap (a_k, b_k)).$$
>
> By subadditivity of measure,

$$\mu(A) \leq \sum_{k=1}^{\infty} \mu(A \cap (a_k, b_k)) \leq (1-\varepsilon) \sum_{k=1}^{\infty} |b_k - a_k| \Rightarrow \frac{\mu(A)}{1-\varepsilon} \leq \sum_{k=1}^{\infty} |b_k - a_k|.$$

By the greatest lower bound definition of infimum,

$$\mu(A) \geq \frac{\mu(A)}{1-\varepsilon} \Rightarrow (1-\varepsilon)\mu(A) > \mu(A)$$

This is a contradiction.

Problem P2

Let $A \subseteq \mathbb{R}$ be measurable with $m(A) > 0$, and let $A - A \coloneqq \{a - a' : a, a' \in A\}$. Prove that $A - A$ contains a nonempty interval around the origin. *Hint (from the sheet):* show there is $\delta > 0$ such that $A \cap (A + r) \neq \emptyset$ for all $|r| < \delta$.

Solution. We will make a preliminary reduction. Suppose we know that there exists a $\delta > 0$ such that for all $r \in (-\delta, \delta)$ we have that $A \cap (A + r) \neq \emptyset$. Then $(-\delta, \delta) \subseteq A - A$, because $\forall r \in (-\delta, \delta)$ there exists $a \in A \cap (A - r) \Rightarrow a \in A$ and $a + r \in A \Rightarrow (a + r) - a = r$. Thus it suffices to prove that there is such a $\delta > 0$.